Soft Entropy Alignment Distillation Method for Mitigating Cross-lingual Attention Behavior Shift in Korean Multilingual Transformers

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Abstract

With the development of large-scale pre-trained language models, multilingual Transformer models have achieved significant performance in natural language processing tasks. Models such as XLM-RoBERTa and mBERT are capable of processing multiple languages using a unified model architecture and tokenizer and are therefore widely applied in cross-lingual natural language understanding tasks. However, these models usually contain a large number of parameters and require high training and deployment costs, making them difficult to use directly in resource-constrained environments. Therefore, knowledge distillation has become an important approach for compressing large-scale Transformer models.

Traditional knowledge distillation mainly guides the learning of the student model through the output logits of the teacher model, enabling the smaller student model to imitate the prediction distribution of the teacher model. However, distillation relying only on output-level information may fail to fully preserve the internal attention behavior of the teacher model. Existing Transformer distillation studies have shown that internal structural information such as hidden states, attention matrices, and self-attention relations also contains important value. For example, TinyBERT incorporates attention matrices and hidden states into distillation objectives, while MiniLM emphasizes that the self-attention module is a key source of knowledge in Transformer distillation. This indicates that internal attention behavior within transformer models should not be ignored.

This paper proposes SEAD, namely Soft Entropy Alignment Distillation. SEAD introduces an attention entropy alignment loss based on Vanilla Knowledge Distillation, allowing the student model to learn not only the output predictions of the teacher model but also its attention entropy behavior. Unlike methods that directly align complete attention maps, SEAD aligns only attention entropy. Therefore, it is a more flexible internal behavior alignment method, particularly suitable for scenarios where the teacher and student models have different architectures.

This paper uses the KLUE-NLI Korean natural language inference task as the primary experimental benchmark. XLM-RoBERTa-base is used as the teacher model, while DistilBERT multilingual cased is used as the student model. Exper-

imental results show that SEAD with β = 0.001 achieves an accuracy of 0.544, which is very close to the 0.547 achieved by Improved Vanilla KD, with only a difference of 0.003. Meanwhile, SEAD reduces the attention entropy distance between the student and teacher models from 33.7223 under Vanilla KD to 11.8484. This indicates that SEAD can significantly improve attention entropy alignment between the student and teacher models while largely maintaining classification performance.

It should be noted that this work is currently limited to a small-scale course project. This paper does not claim that SEAD has completely solved cross-lingual attention behavior shift. Instead, the current experimental results indicate that attention entropy alignment can serve as an effective auxiliary objective in knowledge distillation, enabling the student model to better preserve teacher-like attention behavior while maintaining output performance close to Vanilla KD.

1 Introduction

In recent years, Transformer-based pre-trained language models have become important foundational models in the field of natural language processing. Transformers model dependencies among different tokens within a sequence through the self-attention mechanism. Compared with traditional recurrent neural networks and convolutional neural networks, transformers possess stronger parallel computation capability and long-range dependency modeling ability. Models such as BERT, RoBERTa, XLM-RoBERTa, and DistilBERT are all built upon the Transformer architecture and have been widely applied in tasks including natural language inference, question answering, text classification, and named entity recognition. The success of BERT demonstrated the effectiveness of large-scale pre-trained language models in natural language understanding tasks. BERT learns deep bidirectional contextual representations through pre-training tasks such as Masked Language Modeling, and can then be fine-tuned to various downstream tasks by adding only a simple task-specific output layer. This idea promoted the development of numerous subsequent pre-trained models. Meanwhile, multilingual models gradually became an important direction in cross-lingual NLP research. Models such as mBERT and XLM-RoBERTa can process multiple languages within a unified model, thereby reducing the cost of training separate models for each language. However, large-scale Transformer models usually suffer from problems such as large parameter size, high inference cost, and deployment difficulty. In practical applications, larger computational resources can be used during the training stage, while the deployment stage often requires low latency, low memory consumption, and high inference speed. Therefore, transferring knowledge from large models to smaller models has become one of the core issues in model compression research. Knowledge distillation is a common model compression method. Its basic idea is to use a larger teacher model to guide the learning of a smaller student model. The knowledge distillation work of Hinton et al. pointed out that the soft targets produced by the teacher model contain not only information about the correct category but also relative relationships among categories. By increasing the softmax temperature, smoother probability distributions can be obtained, enabling the student model to learn "dark knowledge" beyond hard labels. Knowledge distillation has been widely applied in Transformer model compression. DistilBERT obtains a smaller and faster model by distilling BERT while retaining most language understanding capability. Patient knowledge distillation argues that the student model should not learn only the final output layer of the teacher model but should gradually learn knowledge from multiple intermediate layers. TinyBERT and MiniLM further demonstrate that internal structures such as hidden states, attention matrices, and self-attention relations can also serve as distillation targets. These studies indicate that traditional logits-level distillation, although effective, may not be sufficient. For Transformers, the attention mechanism is the core component for modeling linguistic relationships. If the student model only learns the final outputs of the teacher model without preserving the teacher model's internal attention behavior, the student model may achieve high accuracy while its internal reasoning process significantly deviates from the teacher. In multilingual scenarios, this problem may become even more complicated. Different languages are tokenized differently under a multilingual tokenizer. Due to complex morphological variations and agglutinative characteristics, Korean may be segmented into more subword tokens under multilingual tokenization. The tokenizer analysis In this paper also shows that Korean exhibits a higher subword-to-word ratio than English and Chinese. This indicates that Korean text may suffer from more severe subword fragmentation under multilingual tokenization. Such fragmentation may affect input sequence length, token representations, and attention allocation, making the student model more likely to exhibit attention behavior shift during distillation. Based on this observation, this paper proposes SEAD (Soft Entropy Alignment Distillation). The core idea of SEAD is to introduce attention entropy alignment loss into traditional knowledge distillation so that the student model learns not only the output logits of the teacher model but also its attention entropy behavior. Attention entropy is used to measure the concentration or dispersion level of attention distributions. Existing studies on Transformer training stability have proposed the concept of attention entropy collapse. For example, Shuangfei Zhai et al., in "Stabilizing Transformer training by Preventing Attention Entropy Collapse," pointed out that abnormally low attention entropy usually indicates excessive concentration of attention probabilities and may be associated with training instability phenomena such as loss oscillation and divergence. Although this study was not specifically designed for Korean multilingual knowledge distillation tasks, it demonstrates that attention entropy can serve as an important indicator for observing Transformer attention behavior. Therefore, this paper adopts this perspective and uses attention entropy to measure internal attention behavior differences between the teacher model and the student model. In this paper, cross-lingual attention behavior shift is mainly used to describe the attention behavior shift phenomenon observed in student models after Korean multilingual distillation. In other words, although the student model may approximate the teacher model in output predictions, significant differences in attention entropy may still exist between them. The objective of SEAD is to reduce this attention entropy gap between the student and teacher through soft entropy alignment loss. The main contributions of this paper are summarized as follows: First, this paper introduces attention entropy as a measure of internal attention behavior differences between teacher and student models in Korean multilingual Transformer distillation scenarios. Second, this paper proposes the SEAD method, which introduces attention entropy alignment loss based on the CE loss and KD loss of Vanilla KD. Third, this paper conducts student fine-tuning, Vanilla KD, Improved Teacher, Improved Vanilla KD, and SEAD experiments on the KLUE-NLI subset. Fourth, through entropy distance to teacher, this paper demonstrates that SEAD can significantly reduce the attention entropy distance between student and teacher models while maintaining accuracy close to Improved Vanilla KD. Fifth, this paper analyzes the effect of entropy alignment loss weights on classification performance through a beta ablation study, demonstrating that β should be carefully selected because excessively large entropy alignment weights may impair classification learning.

2 Related Work

2.1 Transformers and pre-trained language models

Transformer is one of the most important model architectures in the field of natural language processing. It replaces traditional recurrent structures with the self-attention mechanism, enabling the model to directly capture relationships between any two tokens within a sequence. Compared with RNNs, Transformers are more suitable for parallel training and easier to scale to large datasets and large models. BERT performs large-scale pre-training based on the Transformer encoder architecture and learns bidirectional contextual representations through Masked Language Modeling. An important characteristic of BERT is the simplicity of fine-tuning. For different downstream tasks, only a small number of task-specific parameters need to be added to achieve good performance on tasks such as natural language inference, question answering, and text classification. The success of BERT promoted pre-trained language models to become the mainstream paradigm in NLP. However, BERT and subsequent larger Transformer models usually contain a large number of parameters and require high computational costs. Especially in practical deployment scenarios, models need to satisfy requirements related to inference speed, storage space, latency, and energy consumption. Therefore, model compression has become an important issue in the application of pre-trained language models.

2.2 knowledge distillation and Transformer Model Compression

The core idea of knowledge distillation is to transfer knowledge from large models or model ensembles to smaller models. Hinton et al. systematically proposed the concept of distillation, where teacher model soft targets are used to train the student model. Compared with hard labels, soft targets contain richer relational information among categories. For example, in classification tasks, although a category may not be the correct answer, if the teacher model assigns it a relatively high probability, it indicates possible semantic similarity with the correct category. The student model can obtain more refined supervision signals than hard labels by learning such softened distributions. In NLP, DistilBERT is a representative work applying knowledge distillation to pre-trained language model compression. DistilBERT employs knowledge distillation during the pre-training stage and combines language modeling loss, distillation loss, and cosine-distance loss, allowing smaller models to preserve most language understanding capability while reducing model size and inference cost. Patient knowledge distillation further argues that allowing the student model to learn only the final output layer of the teacher model is insufficient. This method enables the student model to learn knowledge from multiple intermediate layers of the teacher model and proposes PKD-Last and PKD-Skip strategies, allowing the student model to gradually absorb information from teacher hidden layers. This idea indicates that intermediate representations of the teacher model contain important knowledge and that attention should not be limited only to final logits. TinyBERT proposes a more comprehensive Transformer distillation method. It aligns not only the logits of the prediction layer, but also the embedding layer, hidden states, and attention matrices. The study of TinyBERT demonstrates that attention matrices and hidden representations within Transformers are valuable distillation targets. MiniLM emphasizes that the self-attention module itself is a key component of Transformers. Instead of performing complex layer-wise alignment, MiniLM uses self-attention distributions and value relations from the final layer of the teacher model to guide student learning. This reduces limitations caused by mismatched teacher and student depths and makes distillation more flexible. These studies collectively indicate that Transformer distillation has evolved from initial output-level logits distillation to internal representation distillation and attention behavior distillation. The SEAD method proposed In this paper belongs to the same research direction but adopts a lighter alignment strategy by aligning attention entropy rather than directly aligning complete attention matrices.

2.3 Multilingual Models and Cross-lingual Representation Learning

Multilingual pre-trained models aim to process multiple languages using a unified model. mBERT is one of the earliest widely used multilingual models and was pre-trained on Wikipedia texts from 104 languages. Research by Pires et al. showed that mBERT possesses a certain degree of zero-shot cross-lingual transfer capability, meaning that a model fine-tuned on one language can be transferred to other languages for evaluation. However, the study also pointed out that the cross-lingual representations of mBERT are not perfect and that systematic differences still exist among languages. XLM-RoBERTa further expanded the scale of multilingual pre-training. XLM-R was trained on over 2 TB of CommonCrawl data covering 100 languages and achieved strong performance in cross-lingual classification, question answering, and named entity recognition tasks. The XLM-R paper also discussed important issues in multilingual models, such as the trade-off between positive transfer and capacity dilution. In other words, when parameters are shared across multiple languages, models may benefit from cross-lingual sharing, but performance for certain languages may also be affected because model capacity is jointly occupied by many languages. XNLI is an important benchmark dataset for cross-lingual natural language inference. It extends the development and test sets of MultiNLI into 15 languages for evaluating cross-lingual sentence understanding. The introduction of XNLI demonstrates that natural language inference is an important task for evaluating cross-lingual understanding capability. This paper mainly uses the KLUE-NLI Korean natural language inference task rather than XNLI. However, studies related to XNLI and XLM-R provide the cross-lingual research background for This paper. Natural language inference is an important task for measuring sentence-level understanding capability, while multilingual models exhibit performance differences across languages. Therefore, studying multilingual Transformer distillation in Korean scenarios is reasonable.

2.4 Attention Head Analysis and Attention Redundancy

The core component of Transformers is multi-head self-attention. In theory, multiple attention heads can capture semantic, syntactic, and positional relationships within input sequences from different perspectives. However, existing studies indicate that the roles of different attention heads are not identical and that not all heads are equally important. Clark et al. conducted a systematic analysis of BERT attention heads and found that different heads exhibit different patterns. For example, some heads tend to focus on special separator tokens, some attend to fixed positional offsets, some broadly attend to entire sentences, while others are associated with syntactic relations or coreference relations. These findings indicate that attention heads indeed contain linguistic structural information, and therefore analyzing attention behavior helps understand the internal mechanisms of Transformers.

Michel et al. raised an important question: Are Sixteen Heads Really Better than One? They found that many attention heads in trained Transformer and BERT models could be removed without significantly degrading model performance. This indicates the existence of attention head redundancy within Transformers. Voita et al. further pointed out that specialized heads often perform more important functions, while many other heads can be pruned. This study indicates that attention heads differ in importance. Some heads may possess stable and interpretable linguistic functions, whereas others contribute less. Abnar and Zuidema reminded researchers that raw attention weights should not be directly interpreted because information is continuously mixed across Transformer layers. Direct observation of raw attention maps from a single layer may fail to accurately reflect the contribution of input tokens to final predictions. Therefore, attention analysis should be conducted carefully. These studies provide two insights for This paper. First, attention behavior is an important clue for understanding Transformer internal mechanisms; therefore, evaluation of distillation performance should not focus solely on final accuracy. Second, raw attention maps continuously mix information across layers, while the teacher and student architectures are not completely identical. Direct alignment of complete attention maps may introduce additional noise. Therefore, this paper uses attention entropy to represent the overall concentration level of attention distributions and uses it to measure attention behavior differences between teacher and student models.

2.5 Attention Entropy and Entropy Collapse

Attention entropy measures the concentration or dispersion level of attention distributions. Given an attention probability distribution, entropy is lower when probability mass is concentrated on a small number of tokens, whereas entropy is higher when the probability distribution is more uniform. Existing studies have discussed attention entropy collapse from the perspective of training stability. Shuangfei Zhai, Tatiana Likhomanenko, and colleagues investigated changes in attention entropy during Transformer training in “Stabilizing Transformer training by Preventing Attention Entropy Collapse.” They found that when attention entropy decreases abnormally, attention distributions tend to become excessively concentrated on a small number of tokens. This phenomenon often co-occurs with training instability, such as training loss oscillation or divergence. The study further analyzed the relationship between attention entropy and the spectral norm of attention logits and proposed σReparam, which mitigates entropy collapse through spectral normalization and learnable scaling of linear layers, thereby improving Transformer training stability. In “Variance Sensitivity Induces Attention Entropy Collapse in Transformers” (2025), Jonghyun Hong and Sungyoon Lee further analyzed potential causes of attention entropy collapse. They pointed out that softmax-based attention is highly sensitive to the variance of attention logits. When differences among attention logits increase, softmax amplifies these differences, causing attention probabilities to become concentrated on individual tokens or a small number of tokens, eventually resulting in attention entropy collapse. The study also observed that decreasing attention entropy increases the norm of the attention probability matrix and further enlarges gradient norms, eventually leading to loss spikes or training instability. Besides the training stability perspective, Attanasio et al. also used attention entropy as an internal behavioral constraint signal in “Entropy-based Attention Regularization Frees Unintended Bias Mitigation from Lists.” Their work addressed unintended bias in hate speech detection and proposed Entropy-based Attention Regularization, which reduces model over-reliance on specific terms by increasing the self-attention entropy of particular tokens. Although this study focuses on bias mitigation rather than Korean multilingual knowledge distillation, it demonstrates that attention entropy can be used not only to analyze model behavior but also as a regularization signal during training. These studies provide important motivation for This paper. This paper does not directly investigate Transformer training divergence issues and does not claim that SEAD completely resolves entropy collapse problems discussed in existing literature. Instead, this paper adopts the usage of attention entropy from these studies and uses attention entropy as an indicator for measuring internal attention behavior differences between the teacher model and the student model. If the student model exhibits large attention entropy differences from the teacher after knowledge distillation, this indicates that the student may imitate the teacher only at the output level while failing to sufficiently preserve the teacher’s internal attention behavior. Therefore, this paper proposes SEAD, which constrains the student model through attention entropy alignment loss, making its attention entropy closer to that of the teacher model.

3 Method

3.1 Preliminary Knowledge: knowledge distillation and Attention Entropy

This paper is based on the knowledge distillation framework. knowledge distillation consists of two models: a teacher model and a student model. The teacher model is usually larger and more powerful, while the student model is smaller and more suitable for deployment. In this paper, XLM-RoBERTa-base is used as the teacher model, while DistilBERT multilingual cased is used as the student model. In standard supervised training, the student model is trained using ground-truth labels and mainly optimizes the cross-entropy loss:

$$L_{CE} = \text{CrossEntropy}(y, p_s)$$

where y denotes the ground-truth labels and p_s denotes the predicted probability distribution of the student model. In knowledge distillation, the student model learns not only the ground-truth labels but also the softened output distribution of the teacher model. Let teacher logits be z_t, student logits be z_s, and temperature be T. The softened probability distributions of the teacher and student can then be expressed as:

$$p_t = \text{softmax}(z_t/T)$$
$$p_s = \text{softmax}(z_s/T)$$

The knowledge distillation loss is usually computed using KL-divergence:

$$L_{KD} = \text{KL}(p_t \| p_s)$$

The overall training objective of Vanilla KD is:

$$L_{VKD} = (1 - \alpha)L_{CE} + \alpha L_{KD}$$

where α controls the balance between ground-truth supervision and teacher soft predictions. In this paper, the temperature T is set to 2.0, while α is set to 0.5. Besides output logits, this paper focuses on internal attention behavior within Transformers. For an attention probability distribution A, attention entropy is defined as:

$$H(A) = -\sum A \log(A)$$

Lower attention entropy indicates that the attention distribution is more concentrated, whereas higher attention entropy indicates that the attention distribution is more dispersed. This paper uses attention entropy as an indicator for measuring internal attention behavior differences between the teacher and student models. It should be emphasized that This paper does not simply interpret attention entropy as "the higher the better" or "the lower the better." More precisely, this paper focuses on whether the student model's attention entropy is close to that of the teacher model. Since the teacher model is a larger model that has been fine-tuned, its attention entropy can be regarded as the reference behavior for student learning. The objective of SEAD is not to blindly increase or decrease entropy but to make the student model's attention entropy closer to that of the teacher model.

3.2 Cross-lingual Attention Behavior Shift

In multilingual Transformer distillation, the student model may approximate the teacher model in output predictions while exhibiting significant differences in internal attention behavior. In other words, the student model may learn the final classification outputs of the teacher model without preserving the teacher model's attention distribution characteristics. In this paper, cross-lingual attention behavior shift is used to describe the internal attention behavior deviation observed in student models after Korean multilingual distillation. Specifically, although the student model may approximate the teacher model in output predictions, significant differences may still exist between its attention entropy and that of the teacher model. This paper adopts the perspective of studies related to attention entropy collapse and considers such differences as reflecting insufficient inheritance of teacher attention behavior by the student model. The objective of SEAD is to reduce the attention entropy gap between the student and teacher models through soft entropy alignment loss. This issue deserves attention in Korean scenarios. Korean exhibits complex morphological variations and agglutinative characteristics and may therefore be segmented into more subword tokens under a multilingual tokenizer. The tokenizer analysis In this paper shows that Korean has a subword-to-word ratio of 2.459798, which is higher than 1.354868 for English and 0.686563 for

Chinese. This indicates that Korean exhibits more obvious token fragmentation under multilingual tokenization. Token fragmentation may bring two effects. First, semantic content of the same length may be segmented into more subword tokens in Korean, thereby increasing sequence processing complexity. Second, a larger number of subword tokens may make attention distributions more complex, causing the student model to have greater difficulty stably imitating teacher attention behavior during distillation. Therefore, in Korean multilingual Transformer distillation, using only logits-level KD may be insufficient. To mitigate attention behavior shift between the student and teacher models, this paper proposes SEAD, which introduces attention entropy alignment loss based on Vanilla KD.

3.3 SEAD: Soft Entropy Alignment Distillation

The full name of SEAD is Soft Entropy Alignment Distillation. The core idea of SEAD is that the student model learns not only the output logits of the teacher model but also its attention entropy behavior. Existing Transformer distillation methods have already demonstrated that internal structural information is important for student learning. TinyBERT transfers multi-layer knowledge from the teacher model to the student model by aligning the embedding layer, hidden states, attention matrices, and prediction logits. MiniLM uses the final-layer self-attention distributions and value relations of the teacher model to guide student learning while avoiding complex layer-wise matching problems. SEAD shares the same viewpoint as these methods in that Transformer distillation should not focus only on output logits. The difference is that SEAD does not directly align complete attention maps but instead aligns attention entropy. This design is motivated by two main reasons. First, the teacher model and student model have different architectures. In this paper, XLM-RoBERTa-base is used as the teacher model and DistilBERT multilingual cased is used as the student model. They differ in model architecture, number of layers, hidden size, and attention heads. If the student model is forced to replicate the complete attention map of the teacher model layer by layer, additional correspondence mappings and dimensional transformations would be required. This would not only increase method complexity but might also introduce instability in small-scale experiments. Second, this paper focuses more on overall changes in attention distributions rather than exact consistency of every attention score. Attention entropy summarizes the concentration level of attention distributions. Lower entropy indicates more concentrated attention on a small number of tokens, whereas higher entropy indicates more dispersed attention. Therefore, SEAD does not require the student model to replicate every attention weight of the teacher model but instead encourages the student model's attention entropy to be close to that of the teacher model. This preserves teacher attention behavior while avoiding overly strong attention map constraints. Specifically, during training, both the teacher model and student model output attention probabilities. Let teacher attention be denoted as $A_{teacher}$ and student attention as $A_{student}$. Their attention entropies are then computed as:

$$H_{teacher} = \text{Entropy}(A_{teacher})$$

$$H_{student} = \text{Entropy}(A_{student})$$

SEAD computes the difference between teacher entropy and student entropy using mean squared error:

$$L_{entropy} = \text{MSE}(H_{teacher}, H_{student})$$

By minimizing $L_{entropy}$, the student model is guided toward an attention entropy distribution closer to that of the teacher model. In other words, SEAD does not force the student model to completely replicate the teacher attention map but instead makes the student model closer to the teacher in terms of the concentration level of attention distributions.

3.4 Training Objective

The training objective of Vanilla KD consists of two components:

$$L_{VKD} = (1 - \alpha)L_{CE} + \alpha L_{KD},$$

where L_{CE} denotes the cross-entropy loss between student predictions and ground-truth labels, L_{KD} denotes the KL-divergence loss between teacher logits and student logits, and α represents the weight of the KD loss.

SEAD introduces attention entropy alignment loss based on Vanilla KD:

$$L_{SEAD} = (1 - \alpha)L_{CE} + \alpha L_{KD} + \beta L_{entropy},$$

where

$$L_{entropy} = \mathrm{MSE}(H_{teacher}, H_{student}).$$

Here, β is a hyperparameter controlling the weight of the entropy alignment loss. If β is too large, the student model may overemphasize attention entropy alignment while neglecting the classification task itself. Conversely, if β is too small, the effect of entropy alignment may become insignificant. Therefore, the selection of β is important.

The final parameters used In this paper are as follows: temperature $T = 2.0$, alpha $\alpha = 0.5$, and final beta $\beta = 0.001$. The selection of $\beta = 0.001$ is based on the results of the beta ablation study. Experimental results show that $\beta = 0.1$ and $\beta = 0.01$ both significantly reduce accuracy, whereas $\beta = 0.001$ improves entropy alignment while maintaining relatively high accuracy.

3.5 Training Procedure

The training procedure of This paper consists of six steps. **Step 1:** Fine-tune the teacher model. XLM-RoBERTa-base is fine-tuned on the KLUE-NLI subset to obtain Improved Teacher XLM-R.
Step 2: Train the student baseline. DistilBERT multilingual cased is directly trained on the KLUE-NLI subset using ground-truth labels as the Student Fine-tuning baseline.
Step 3: Train the Vanilla KD baseline. Teacher logits are used to guide student learning, and the training objective includes CE loss and KD loss.
Step 4: Train Improved Vanilla KD. The fine-tuned improved teacher is used as the teacher model to make the KD baseline more reliable.
Step 5: Train the SEAD student. Attention entropy alignment loss is introduced based on Improved Vanilla KD, enabling the student model to learn both teacher output predictions and attention entropy behavior.
Step 6: Conduct result analysis. Accuracy is used to evaluate classification performance, while attention entropy distance to teacher is used to evaluate differences in internal attention behavior between the student and teacher models.

4 Experimental Design

4.1 Dataset

This paper mainly uses the KLUE-NLI dataset for experiments. KLUE-NLI is a Korean natural language inference task whose objective is to determine the logical relationship between two sentences. The task usually contains three labels: entailment, neutral, and contradiction. Due to limited computational resources, this paper does not use the full KLUE-NLI dataset for large-scale training but instead conducts experiments on a subset. The Improved Teacher and main student experiments use 5,000 training samples and 1,000 validation samples. In addition, this paper performs tokenizer fragmentation analysis, computing subword-to-word ratios for English, Chinese, and Korean texts to analyze segmentation differences of the multilingual tokenizer across different languages.

4.2 Model Settings

The teacher model used In this paper is XLM-RoBERTa-base. XLM-R is selected as the teacher model because it is a representative multilingual Transformer model capable of processing multiple languages, including Korean. Studies related to XLM-R indicate that large-scale multilingual pre-training can improve performance on cross-lingual understanding tasks, although multilingual models also suffer from issues such as capacity dilution. The student model used In this paper is DistilBERT multilingual cased. DistilBERT multilingual cased is selected as the student model because it is lighter than large teacher models and better aligns with the objective of model compression in knowledge distillation.

4.3 Comparison Methods

This paper compares the following methods. Student Fine-tuning: DistilBERT multilingual cased is directly trained using ground-truth labels without using a teacher model. Vanilla KD: Teacher logits are used to guide student learning. However, since the early teacher model was not sufficiently

Method	Accuracy
Student Fine-tuning	0.352
Vanilla KD	0.372
Improved Teacher XLM-R	0.499
Improved Vanilla KD	0.547
SEAD $\beta = 0.001$	0.544

Table 1: Comparison of accuracy results in the main experiment.

fine-tuned, this result is mainly used to validate the KD pipeline. Improved Teacher XLM-R: XLM-RoBERTa-base is fine-tuned on the KLUE-NLI subset and used as a more reliable teacher model. Improved Vanilla KD: Vanilla KD is performed using the improved teacher, serving as the primary comparison method for SEAD. SEAD β = 0.001: Attention entropy alignment loss is introduced based on Improved Vanilla KD. This is the method proposed In this paper.

4.4 Evaluation Metrics

This paper uses two main evaluation metrics. The first metric is accuracy. Accuracy is used to measure classification performance on the KLUE-NLI validation set. The second metric is attention entropy distance to teacher. This metric measures the distance between the attention entropy of the student model and that of the teacher model. A smaller distance indicates that the student model's attention entropy behavior is closer to that of the teacher model. Attention entropy distance is defined as:

$$Distance = |H_{student} - H_{teacher}|$$

where $H_{student}$ denotes the average attention entropy of the student model, and $H_{teacher}$ denotes the average attention entropy of the teacher model.

5 Experimental Results

5.1 Main Experimental Results

The main experimental results are shown in Table 1.

As shown in Table 1, the accuracy of Student Fine-tuning is 0.352, which is only slightly higher than the random baseline for three-class classification. The accuracy of Vanilla KD is 0.372, showing some improvement compared with the student baseline. However, since the early teacher model was not sufficiently fine-tuned, the improvement remains limited. After fine-tuning on the KLUE-NLI subset, the accuracy of Improved Teacher XLM-R reaches 0.499. After performing distillation using the improved teacher, the accuracy of Improved Vanilla KD further increases to 0.547, which is the highest result in the current experiments. The accuracy of SEAD β = 0.001 is 0.544, which is very close to the 0.547 achieved by Improved Vanilla KD, with only a difference of 0.003. This indicates that introducing attention entropy alignment loss does not significantly impair the classification performance of the student model. It should be emphasized that SEAD does not outperform Improved Vanilla KD in the current experiments. Therefore, this paper does not claim that SEAD is superior to Vanilla KD in terms of accuracy. A more accurate conclusion is that SEAD further improves attention entropy alignment between the student and teacher models while maintaining comparable classification accuracy.

5.2 Attention Entropy Analysis

The results of the attention entropy analysis are shown in Table 2.

As shown in Table 2 and Figure 1, the average attention entropy of Improved Teacher XLM-R is 156.4763. The average attention entropy of Improved Vanilla KD Student is 190.1986, and its entropy distance to teacher is 33.7223. This indicates that although Vanilla KD achieves relatively high accuracy, there are still significant differences between the student model and the teacher model in terms of internal attention entropy behavior. In contrast, the average attention entropy of SEAD β = 0.001 Student is 168.3247, and its entropy distance to teacher is 11.8484, which is significantly lower

Model	Attention Entropy	Entropy Distance to Teacher
Improved Teacher XLM-R	156.4763	0.0000
Improved Vanilla KD Student	190.1986	33.7223
SEAD $\beta = 0.001$ Student	168.3247	11.8484

Table 2: Results of attention entropy analysis.

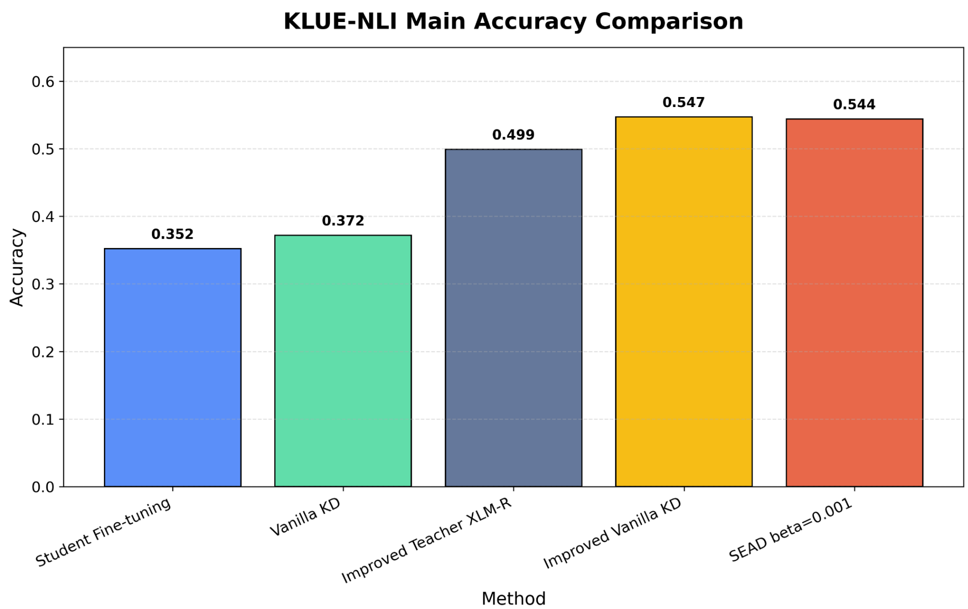

Figure 1: Comparison of accuracy results in the main experiment.

than the 33.7223 of Improved Vanilla KD. This indicates that SEAD can make the student model's attention entropy closer to that of the teacher model. This result supports the core hypothesis of This paper: using only output-level KD may be insufficient to preserve the internal attention behavior of the teacher model, while SEAD can improve internal behavior alignment between the student and teacher models by introducing attention entropy alignment loss.

5.3 Beta Ablation Study

To analyze the effect of entropy alignment loss weight on performance, this paper conducts experiments with different beta values. The results are shown in Table 3.

As shown in Figure 2, when β = 0.1, the accuracy of SEAD is only 0.360, which is close to the student baseline. This indicates that excessively large entropy alignment weights significantly affect classification learning. When β = 0.01, the accuracy increases to 0.417, but it is still lower than Improved Vanilla KD. When β = 0.001, the accuracy reaches 0.544, which is very close to Improved Vanilla KD. This indicates that entropy alignment loss should serve as an auxiliary objective rather than the dominant training objective. If β is too large, the student model may overemphasize attention entropy alignment, thereby weakening the contributions of CE loss and KD loss to the classification task. An appropriate beta value can improve entropy alignment while maintaining classification performance. Therefore, this paper finally selects β = 0.001 as the primary setting for SEAD.

5.4 Tokenizer Fragmentation Analysis

This paper also analyzes the tokenization behavior of the multilingual tokenizer across different languages. The results are shown in Figure 3.

As shown in Figure 3, Korean has a subword-to-word ratio of 2.459798, which is higher than 1.354868 for English and 0.686563 for Chinese. This indicates that Korean text tends to be segmented into more

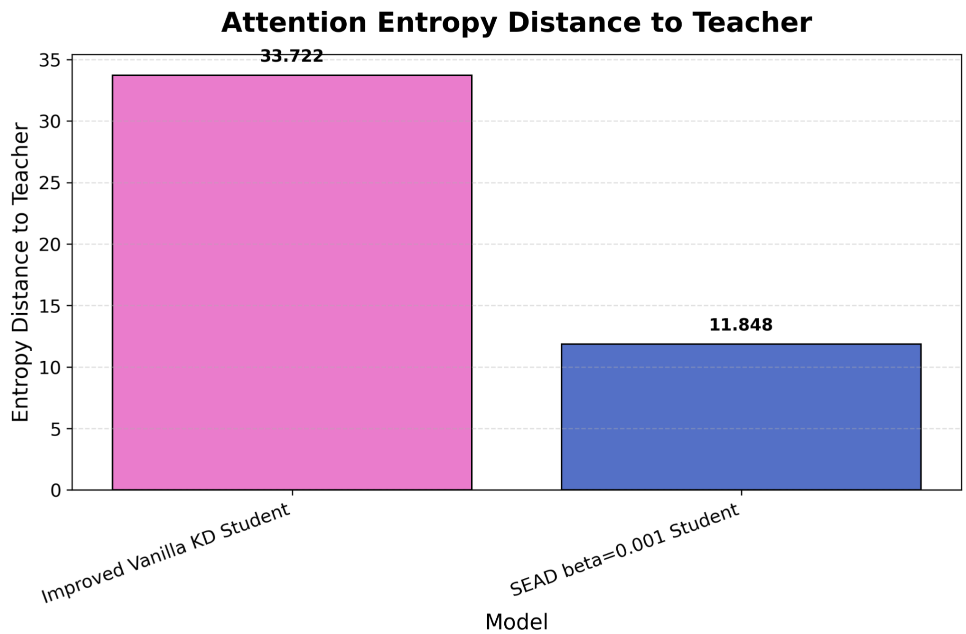

Figure 2: Comparison of Attention Entropy Distance Between Student and Teacher

subword tokens under a multilingual tokenizer, exhibiting more obvious tokenizer fragmentation. This result provides motivation for studying Korean multilingual Transformer distillation In this paper. Since Korean exhibits more severe subword fragmentation under multilingual tokenization, the student model may be more likely to exhibit attention behavior shift during distillation. Therefore, introducing attention entropy alignment is reasonable. It should be noted that the word counting method for Chinese differs from that of English and Korean. Therefore, the Chinese results should not be overinterpreted. This paper mainly focuses on the higher subword-to-word ratio observed in Korean relative to English.

6 Analysis and Discussion

6.1 Effect of SEAD on Classification Performance

According to the main experimental results, SEAD $\beta = 0.001$ achieves an accuracy of 0.544, while Improved Vanilla KD achieves 0.547. Although SEAD does not outperform Improved Vanilla KD, the difference is very small, only 0.003. This indicates that introducing attention entropy alignment does not significantly impair the classification capability of the student model. In knowledge distillation tasks, this is an important result because introducing additional internal alignment losses may lead to conflicts among training objectives. If the entropy alignment loss becomes too strong, the student model may fail to learn the classification task effectively. The beta ablation study also supports this observation: when $\beta = 0.1$, model accuracy decreases significantly. Therefore, the value of SEAD does not lie in simply improving accuracy but rather in improving internal attention behavior alignment between teacher and student models while maintaining comparable accuracy.

6.2 Effect of SEAD on Attention Entropy

The attention entropy analysis represents the most important experimental result of This paper. The entropy distance to teacher of Improved Vanilla KD is 33.7223, whereas that of SEAD $\beta = 0.001$ is 11.8484. Compared with Vanilla KD, SEAD significantly reduces the attention entropy difference between the student and teacher models. This indicates that although Vanilla KD improves classification performance through logits transfer, it cannot guarantee that the internal attention behavior of the student model remains consistent with that of the teacher model. By introducing attention entropy alignment loss, SEAD constrains the student model using teacher attention entropy

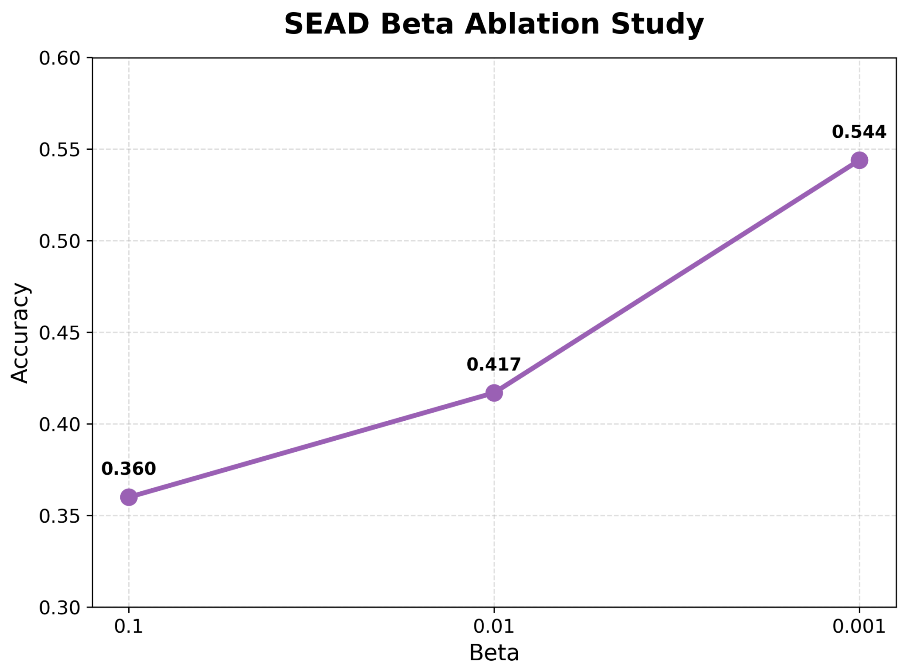

Figure 3: Results of the SEAD Beta Ablation Study

while learning output distributions, thereby preserving more teacher-like internal behavior. This result suggests that evaluating only accuracy may be insufficient during model compression. Even if the student model achieves classification results close to those of the teacher model, it does not necessarily imply that their internal attention behaviors are identical. In the experiments of this paper, Improved Vanilla KD achieves slightly higher accuracy than SEAD but exhibits a larger attention entropy gap from the teacher model. In contrast, SEAD maintains comparable accuracy while making student attention entropy closer to the teacher model. This indicates that attention entropy alignment can serve as a complementary constraint beyond logits distillation, helping the student model better preserve the teacher's internal attention behavior.

6.3 Why Not Directly Align Attention Maps

Direct alignment of attention maps between teacher and student models is a possible approach, and studies such as TinyBERT have indeed demonstrated that attention matrix distillation can effectively improve student learning performance. However, directly aligning complete attention maps may not be appropriate In this paper because the teacher and student models have different architectures. XLM-RoBERTa-base and DistilBERT multilingual cased may differ in terms of layer numbers, hidden size, and attention heads. If attention maps are forcibly aligned layer by layer and head by head, complex layer mappings and dimensional transformations may be required. The idea of MiniLM provides important inspiration for This paper. MiniLM uses final-layer self-attention knowledge from the teacher model to guide student learning, avoiding limitations of complex layer-to-layer distillation. SEAD further adopts a lighter approach by aligning only attention entropy rather than complete attention distributions or value relations. Therefore, the advantages of SEAD lie in its simplicity, flexibility, and better compatibility with teacher–student architectural differences. It does not require the student model to replicate every attention score of the teacher model but instead encourages the student model to approximate the teacher model in terms of overall attention distribution concentration.

6.4 Significance of Attention Entropy

Existing studies on attention entropy collapse indicate that attention entropy is related to Transformer training stability. Zhai et al. found that abnormally low attention entropy often co-occurs with training

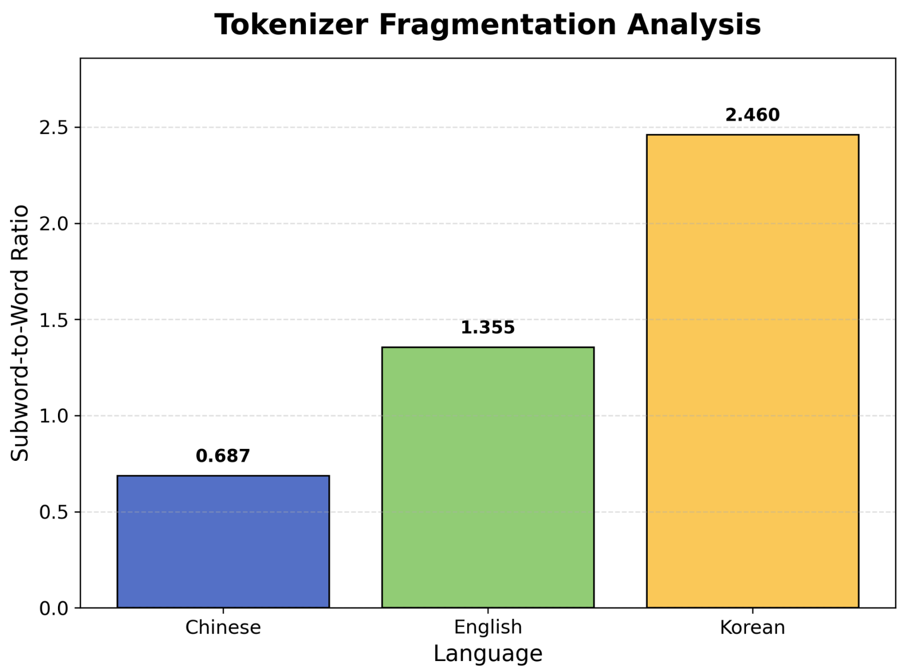

Figure 4: Comparison of Multilingual Tokenizer Fragmentation Results

instability. Research on variance sensitivity in 2025 further showed that softmax amplifies differences among attention logits, causing attention probabilities to become overly concentrated and leading to entropy collapse and gradient instability. This paper does not directly investigate training instability and does not demonstrate that shifts in student attention entropy necessarily cause training divergence. Therefore, this paper cannot directly claim that SEAD solves training stability problems. However, these studies demonstrate that attention entropy is a meaningful internal behavior indicator because it reflects the concentration or dispersion level of attention distributions. Therefore, using attention entropy as a teacher–student internal behavior alignment signal is reasonable. Experimental results of This paper also show that SEAD significantly reduces entropy distance between teacher and student models, indicating that attention entropy alignment has practical value in knowledge distillation.

6.5 Effect of Beta

Beta is an important hyperparameter in SEAD because it determines the weight of entropy alignment loss within the total loss. Experimental results show that $\beta = 0.1$ and $\beta = 0.01$ significantly reduce accuracy, whereas $\beta = 0.001$ achieves a better balance. This indicates that attention entropy alignment should not be overly strong. Excessive alignment forces the student model to overemphasize consistency of attention entropy with the teacher model while ignoring classification information from ground-truth labels and teacher logits. Therefore, entropy alignment should act as an auxiliary objective rather than dominating the training process. Consequently, this paper selects $\beta = 0.001$ as the trade-off between accuracy and attention entropy alignment.

7 Conclusion

This paper proposed SEAD, a Soft Entropy Alignment Distillation method that adds attention entropy alignment loss to Vanilla Knowledge Distillation. The experimental results show that SEAD with $\beta = 0.001$ achieves accuracy close to Improved Vanilla KD while reducing the entropy distance between the student and teacher models. This suggests that attention entropy can be used as a useful auxiliary signal for preserving teacher-like internal attention behavior in Korean multilingual Transformer distillation. However, this paper is still a small-scale course experiment, and future work should evaluate SEAD on larger datasets, more language settings, and different teacher–student architectures.

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A Appendix

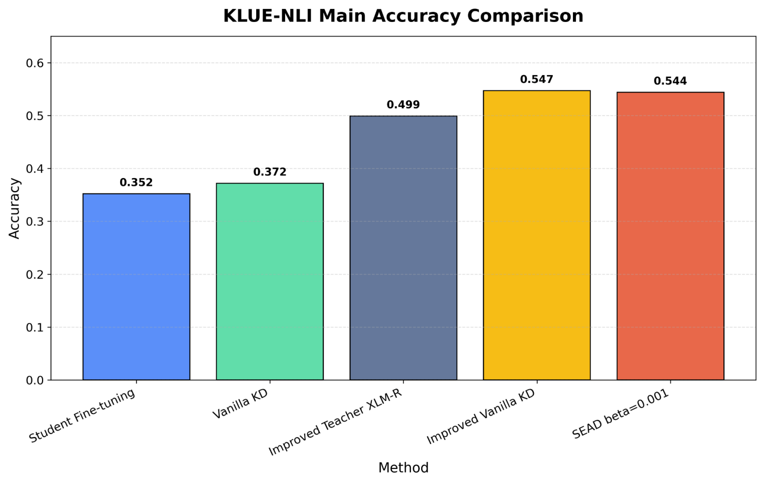

Figure 5: Comparison of accuracy results in the main experiment.

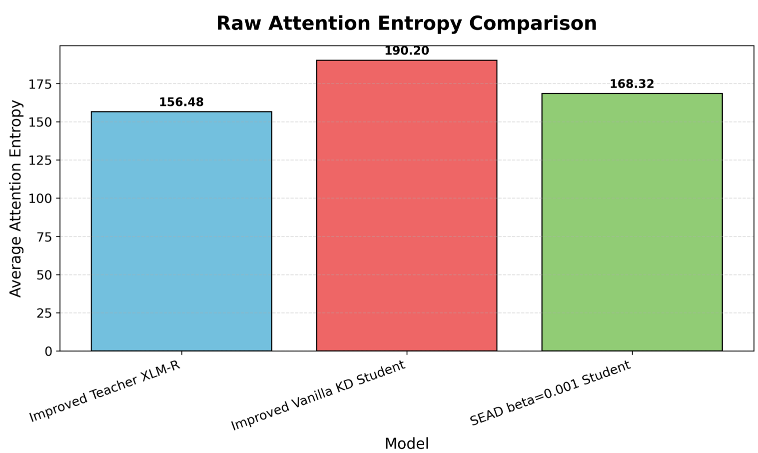

Figure 6: Comparison of average attention entropy results.